INCH-POUND

MIL-M-28787/274B(SH)
AMENDMENT 1
29 March 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TIMER, DIGITAL, KEY CODE YKV

This amendment forms a part of Military Specification MIL-M-28787/274B(EC), dated 22 August 1983, and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 5

TABLE I, Test conditions P1, P2, 25 ± 5°C, Min column: Delete "72.9" and substitute "71.35".

TABLE I, Test conditions P1, P2, Class I, 0 ± 5°C, Min column: Delete "72.7" and substitute "71.15".

TABLE I, Test conditions P1, P2, Class I, 60 ± 5°C, Min column: Delete "73.2" and substitute "71.65".

TABLE I, Test conditions P1, P2, Class II, -55 ± 5°C, Min column: Delete "72.3" and substitute "70.75".

TABLE I, Test conditions P1, P2, Class II, 85 ± 5°C, Min column: Delete "73.3" and substitute "71.75".

TABLE I, Test conditions R1, R2, 25 ± 5°C, Max column: Delete "24.6" and substitute "26.0".

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TABLE I, Test conditions R1, R2, Class I, 0 ± 5°C, Max column: Delete "24.5" and substitute "25.9".

TABLE I, Test conditions R1, R2, Class I, 60 ± 5°C, Max column: Delete "24.7" and substitute "26.1".

TABLE I, Test conditions R1, R2, Class II, -55 ± 5°C, Max column: Delete "24.1" and substitute "25.5".

TABLE I, Test conditions R1, R2, Class II, 85 ± 5°C, Max column: Delete "24.8" and substitute "26.2".

TABLE I, Test conditions S1, S2, 25 ± 5°C, Max column: Delete "25.1" and substitute "26.5".

TABLE I, Test conditions S1, S2, Class I, 0 ± 5°C, Max column: Delete "25.0" and substitute "26.4".

TABLE I, Test conditions S1, S2, Class I, 60 ± 5°C, Max column: Delete "25.2" and substitute "26.6".

TABLE I, Test conditions S1, S2, Class II, -55 ± 5°C, Max column: Delete "24.9" and substitute "26.3".

TABLE I, Test conditions S1, S2, Class II, 85 ± 5°C, Max column: Delete "25.3" and substitute "26.8".

PAGE 6

TABLE I, Test conditions T10, T22, 25 ± 5°C, Max column: Delete "5.86" and substitute "5.9".

TABLE I, Test conditions T10, T22, Class I, 0 ± 5°C, Max column: Delete "6.66" and substitute "6.7".

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TABLE I, Test conditions T10, T22, Class I, 60 ± 5°C, Max column: Delete "6.70" and substitute "6.74".

TABLE I, Test conditions T10, T22, Class II, -55 ± 5°C, Max column: Delete "6.66" and substitute "6.7".

TABLE I, Test conditions T10, T22, Class II, 85 ± 5°C, Max column: Delete "6.72" and substitute "6.76".

TABLE I, Test conditions T11, T23, 25 ± 5°C, Max column: Delete "29.2" and substitute "29.5".

TABLE I, Test conditions T11, T23, Class I, 0 ± 5°C, Max column: Delete "33.25" and substitute "33.55".

TABLE I, Test conditions T11, T23, Class I, 60 ± 5°C, Max column: Delete "33.4" and substitute "33.7".

TABLE I, Test conditions T11, T23, Class II, 85 ± 5°C, Max column: Delete "33.5" and substitute "33.8".

TABLE I, Test conditions T12, T24, 25 ± 5°C, Max column: Delete "58.6" and substitute "59.0".

TABLE I, Test conditions T12, T24, Class I, 0 ± 5°C, Max column: Delete "66.6" and substitute "67.0".

TABLE I, Test conditions T12, T24, Class I, 60 ± 5°C, Max column: Delete "67.0" and substitute "67.4".

TABLE I, Test conditions T12, T24, Class II, -55 ± 5°C, Max column: Delete "66.6" and substitute "67.0".

TABLE I, Test conditions T12, T24, Class II, 85 ± 5°C, Max column: Delete "67.2" and substitute "67.6".

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PAGE 8

TABLE I, Test condition CC, Max columns: Delete "400" (five places) and substitute "1500" (five places).

PAGE 10

4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. Test conditions A and B are stress tests and shall be performed only at 25 ± 5°C. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A and B where applicable, followed by the remaining tests in the sequence."

PAGE 18

FIGURE 7: Delete and substitute new figure 7 as printed on page 7 of this amendment.

PAGE 21

FIGURE 9: Delete and substitute new figure 9 as printed on page 8 of this amendment.

PAGE 22

FIGURE 10: Delete and substitute new figure 10 as printed on page 9 of this amendment.

PAGE 31

TABLE XV, Test conditions P1, P2, Min columns: Delete "72.9" (two places) and substitute "71.35" (two places).

PAGE 35

TABLE XVII, Test conditions R1, R2, Max columns: Delete "24.6" (two places) and substitute "26.0" (two places).

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TABLE XVIII, Test conditions S1 and S2, Max columns: Delete "25.1" (two places) and substitute "26.5" (two places).

PAGE 38

TABLE XIX, Test condition T10, Max column: Delete "5.86" (one place) and substitute "5.9" (one place).

TABLE XIX, Test condition T11, Max column: Delete "29.2" (one place) and substitute "29.5" (one place).

TABLE XIX, Test condition T12, Max column: Delete "58.6" (one place) and substitute "59.0" (one place).

PAGE 39

TABLE XIX, Test condition T22, Max column: Delete "5.86" (one place) and substitute "5.9" (one place).

TABLE XIX, Test condition T23, Max column: Delete "29.2" (one place) and substitute "29.5" (one place).

TABLE XIX, Test condition T24, Max column: Delete "58.6" (one place) and substitute "59.0" (one place).

PAGE 50

TABLE XXVIII, Test condition CC, Max column: Delete "400" (one place) and substitute "1500" (one place).

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PAGE 54

FIGURE 27: Delete and substitute new figure 27 as printed on page 10 of this amendment.

PAGE 63

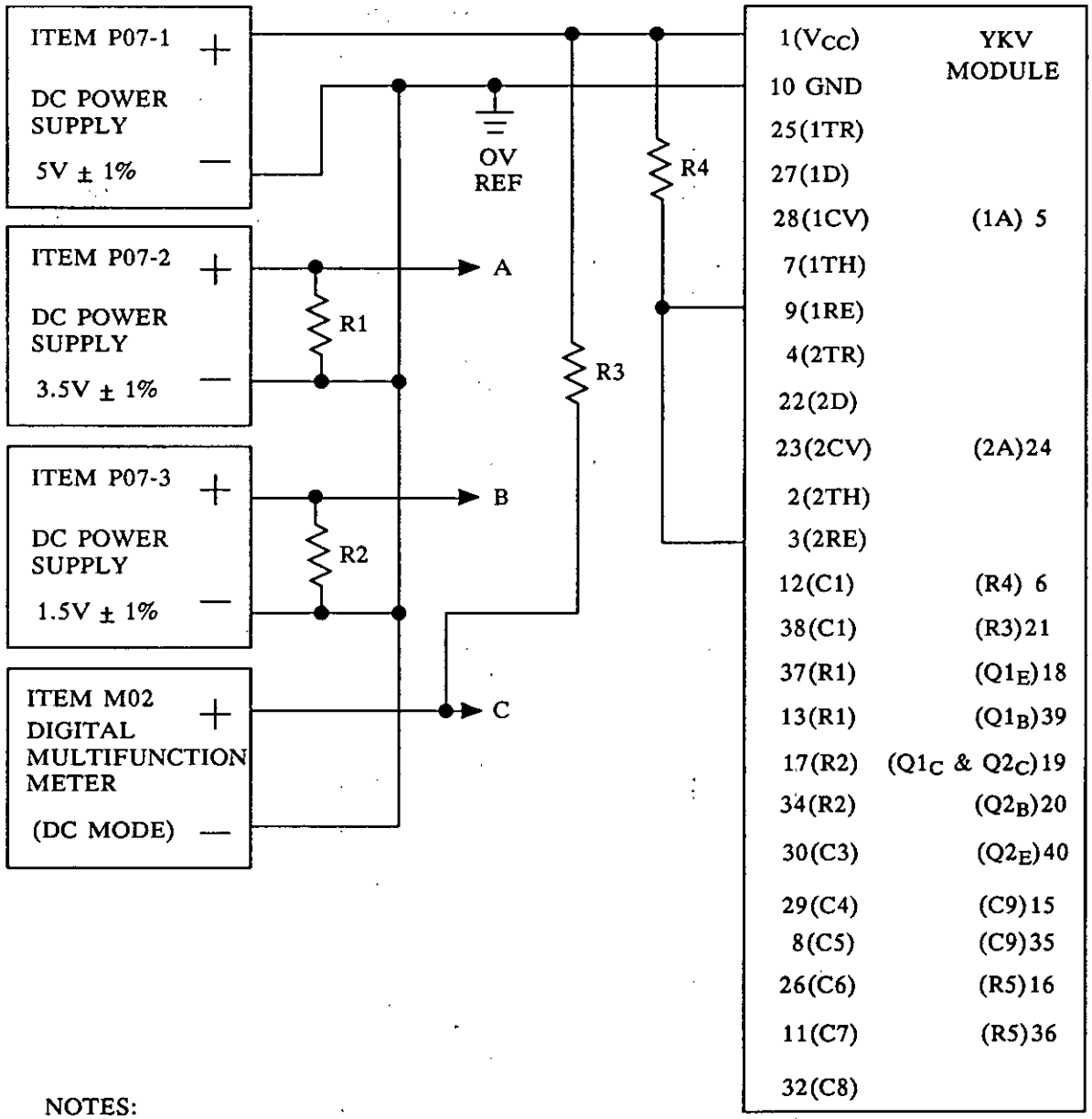
FIGURE 36: Delete and substitute new figure 36 as printed on page 11 of this amendment.

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Agent: NW

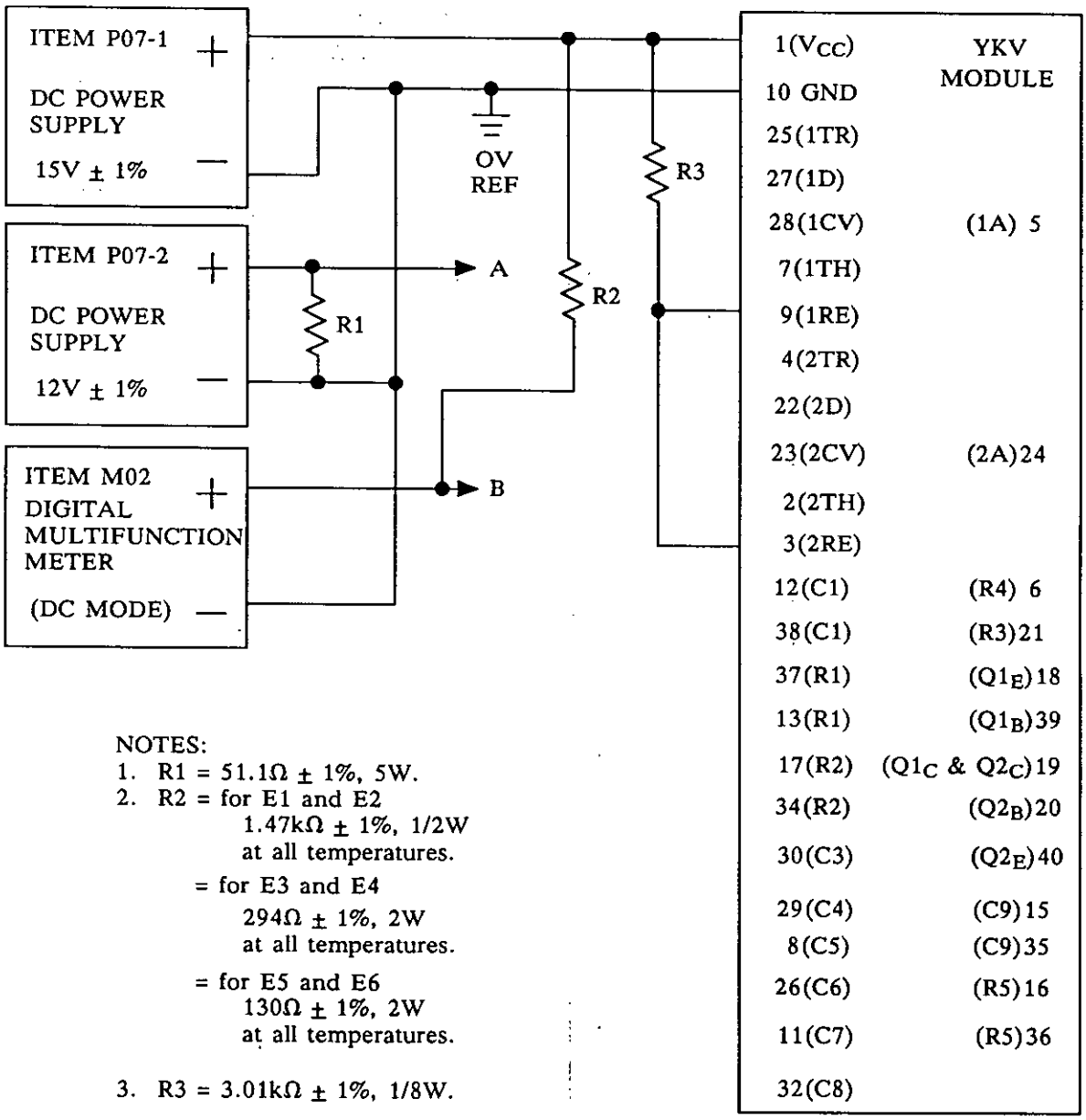
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NOTES:
1. R1 = R2 = 51.1Ω ± 1%, 1/2W.
2. R3 = 953Ω ± 1%, 1/8W @ 25°C.
   931Ω ± 1%, 1/8W @ -55°C, 0°C, 60°C, 85°C.
3. R4 = 3.01kΩ ± 1%, 1/8W.

FIGURE 7. Output voltage (low-level) test circuit for table IV.

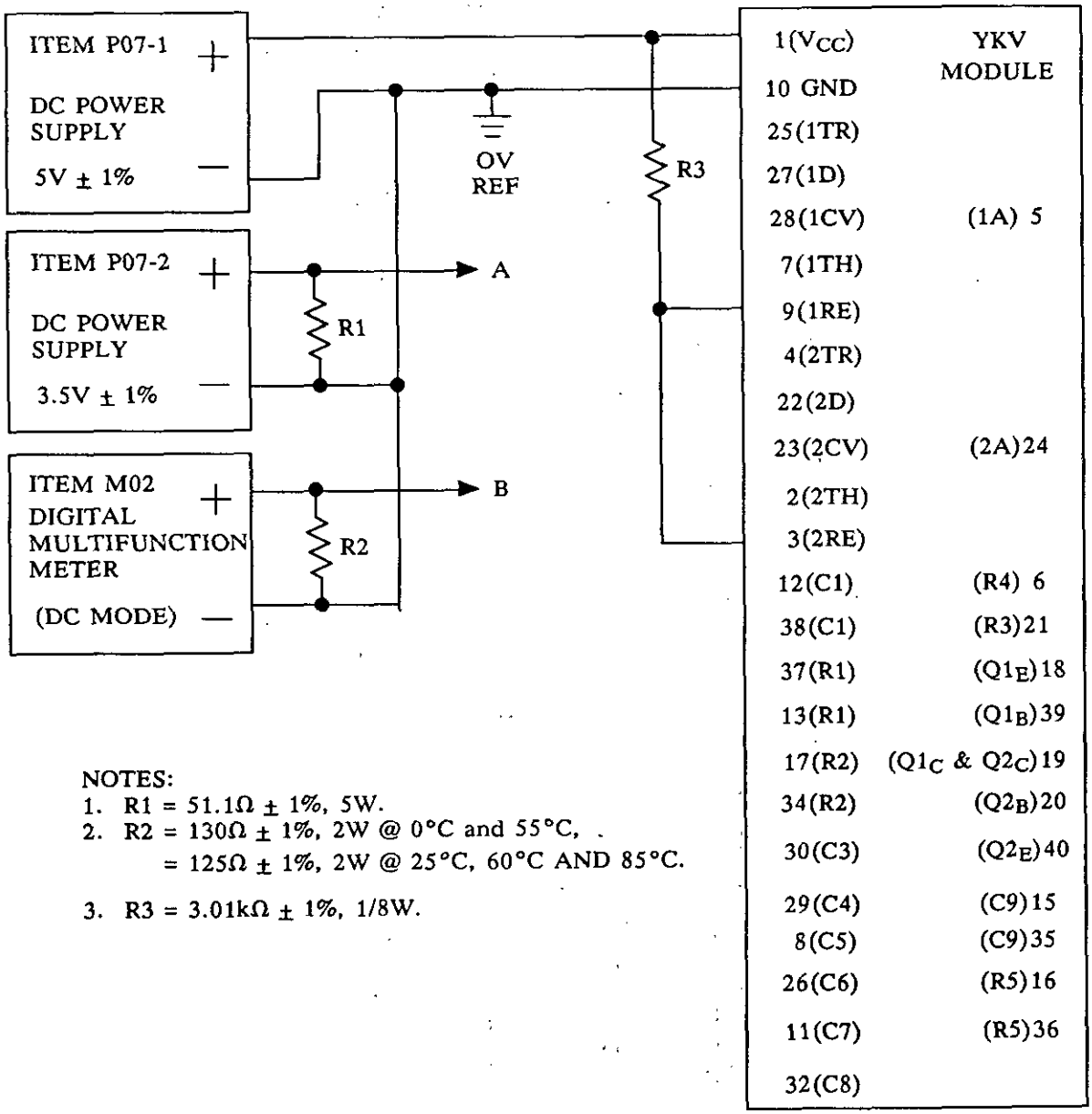
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NOTES:
1. R1 = 51.1Ω ± 1%, 5W.
2. R2 = for E1 and E2
   1.47kΩ ± 1%, 1/2W
   at all temperatures.
   = for E3 and E4
   294Ω ± 1%, 2W
   at all temperatures.
   = for E5 and E6
   130Ω ± 1%, 2W
   at all temperatures.
3. R3 = 3.01kΩ ± 1%, 1/8W.

FIGURE 9. Output voltage (low-level) test circuit for table VI.

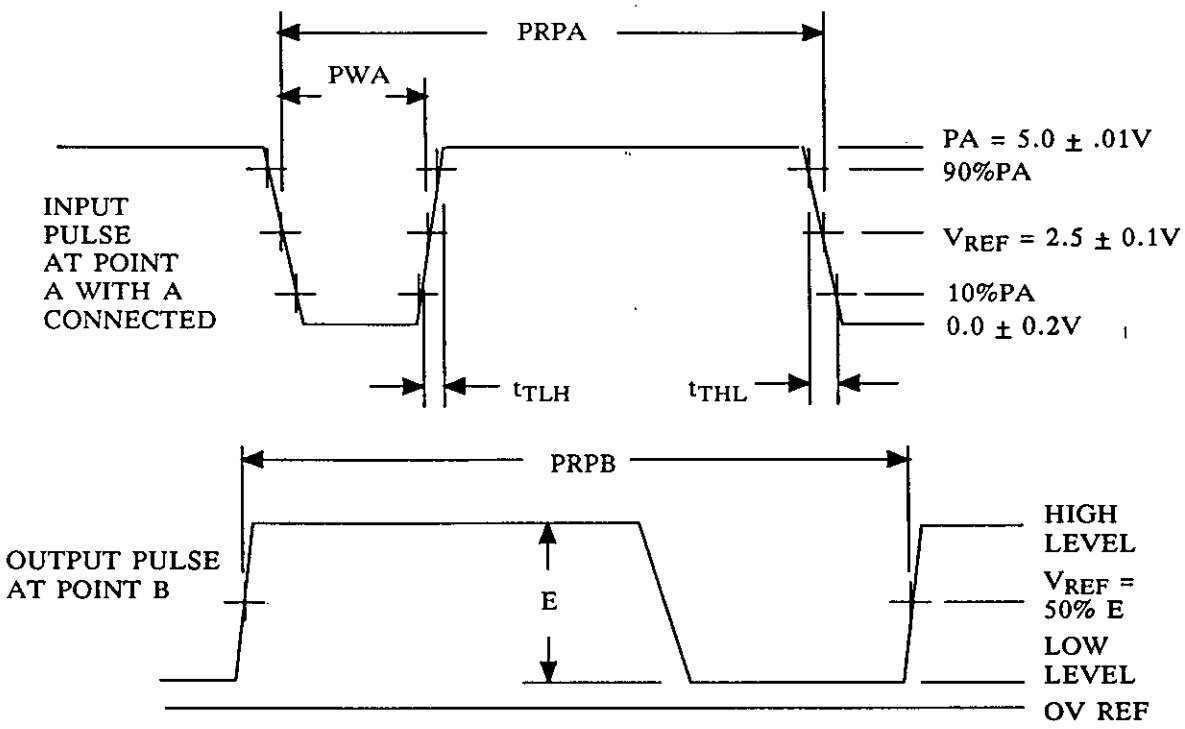
Source: https://assist.dla.mil -- Downloaded: 202610-20T17:19Z
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NOTES:
1. R1 = 51.1Ω ± 1%, 5W.
2. R2 = 130Ω ± 1%, 2W @ 0°C and 55°C,
   = 125Ω ± 1%, 2W @ 25°C, 60°C AND 85°C.
3. R3 = 3.01kΩ ± 1%, 1/8W.

FIGURE 10. Output voltage (high-level) test circuit for table VII.

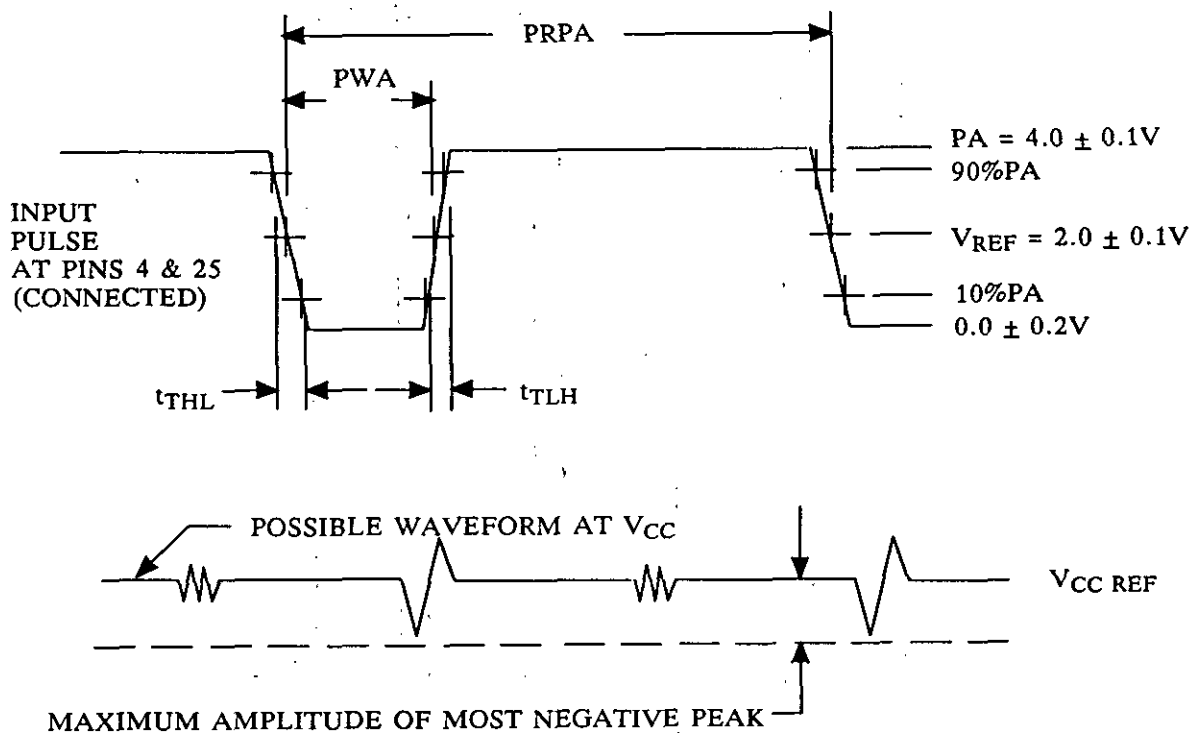
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NOTES;
1. Input pulse parameters for item J01:
   PRPA = 13.7 ± 0.2µs.
   PWA = 200 ± 10ns.
   $t_{TLH} = t_{THL}$ = 10 ± 2ns.
2. Verify that all inputs have a high level ≥ 3V and a low level ≤ 0.25V.
3. The physical length of electrical conductors from items Y01-1 and Y01-2 to input and output pins shall be less than 1 inch (25.4mm).
4. All voltages are referenced to ground.

FIGURE 27. Maximum frequency and minimum trigger pulse check test circuit waveforms for table XXI.

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NOTES;
1. Input pulse parameters for item J01:
   PRPA = 13.7µs ± 10%.
   PWA = 1.1µs ± 10%.
   $t_{TLH} = t_{THL} = 10 \pm 2$ns.
2. Verify the input pulse parameters at each input pin before making measurements.
3. Verify the output voltages low levels and high levels are ≤ 0.25V and ≥ 3.2V respectively.
4. All voltages are referenced to ground.

FIGURE 36. Power supply transient amplitude test circuit waveforms for table XXVIII.

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